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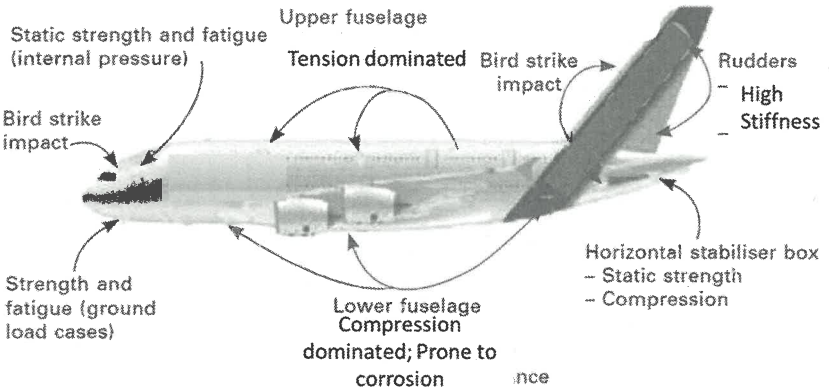
FIRST SEMESTER M.Tech EXAMINATION, JANUARY-2025

AIRCRAFT MATERIALS

Time: 3 Hours **Maximum Marks: 100**

Instructions:
- **i.** Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Then solve Q3 or Q4, Q5 or Q6, Q7 or Q8. Each question carries 20 marks.
- **ii.** Missing data may be suitably assumed.

		Questions	CO's	Blooms Level	Marks
1.	**a.**	A sketch of a large commercial transport aircraft is shown in the figure below. The major load pattern on various portions are also indicated.	**1**	**3**	



Among the available candidate materials as given below, recommend a material for each of the following parts of the structure. Justify your choice with the most important factor(s) in making the choice.

Parts	Materials available	Marks
i. Upper fuselage skins	**i.** Ti-alloy Ti6Al4V	**2**
ii. Lower fuselage skins	**ii.** Maraging steel (300) 18 Ni	**2**
iii. Fin and Wing leading edges	**iii.** Low Alloy steel 300M	**2**
	iv. Al-alloy 2024 (in various tempers)	
iv. Windshield	**v.** Al-alloy 7075 (in various tempers)	**1**
v. Passenger cabin window	**vi.** Honeycomb Sandwich with carbon/epoxy face-sheets	**1**
vi. Rudder	**vii.** Stretched Acrylic	**2**
	viii. Polycarbonate	

	b.	Crashes of Comet and B47 aircraft in 1950's are regarded as "Landmark Events" in the evolution of aircraft structures. Explain with reasoning the	**1**	**2**	**10**

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		major changes that these crashes brought out and the major aspects of material behaviour that were introduced in the design of aircraft structures.			
2.	**a.**	Draw **on a single plot**, schematic stress-strain curves for the following: **i.** Typical Al-alloy used in aircraft **ii.** Low -Alloy steel **iii.** E-Glass fibre **iv.** High-strength (HS) carbon fibre, typically T300 **v.** Unidirectional carbon fibre composite T300/epoxy **vi.** Typical epoxy resin Describe the essential characteristics of each material as you see from the stress-strain curves	**2**	**3**	**10**
	b.	A wing panel of an aircraft may be considered as a flat wide plate for design calculations here. The design allowable stress for the panel is arrived at by considering a knock-down factor of 1.5 over the A-basis yield strength of the material. The panel is to be fabricated from a Ti-alloy that has a plane strain fracture toughness of 55 MPa-m$^{1/2}$. The tension tests using 12 specimens of this material showed mean yield strength of 910 MPa with a standard deviation of 16 MPa. Determine the flaw size resolution limit that the inspection apparatus should have in order to detect a critical flaw in this panel? ***Hints**: (1) The Stress intensity Factor for a crack in a wide panel can be taken as $K=\sigma\sqrt{(\pi a)}$, where a =semi-crack-length.* *(2) Following statistical data is available, where n=number of specimens, K_A = A-basis factor.*	**2**	**3**	**10**

n	7	8	9	10	11	12	13	14	15	16
K_A	4.642	4.354	4.143	3.981	3.852	3.747	3.659	3.585	3.52	3.464

3.	**a.**	Answer following questions in brief. Each question carries 2 marks **i.** A copper wire is drawn from a raw stock of 5 mm diameter to have a diameter of 4.5 mm. How much cold work has the wire undergone? **ii.** What is the correct sequence of phenomena happening during age-hardening treatment of Al-alloys? **iii.** For artificial ageing of Al-alloys, state the effect of higher ageing temperature on peak strength achieved and the ageing time **iv.** Carbon is an important alloying element in iron to make steel. Why	**2**	**2**	**2** **2** **2** **2**

		the solid solution strengthening is very effective with carbon solute in iron? **v.** A metal element A forms a substitutional solid solution with a metal element B. State the conditions under which we may have a full range of compositions of the solid solutions?			**2**
	b.	Differentiate between the following: **i.** Diffusion transformations (such as in precipitation hardening) and Diffusionless transformations (such as martensitic transformations in steel) **ii.** Phase diagrams and TTT diagrams **iii.** Al-Cu alloys and Al-Zn Alloys	**3**	**2**	 **4** **3** **3**
		OR			
4.	**a.**	An alloy has an average grain diameter of 5.76×10^{-2}mm. When tested, it showed the yield strength to be 140 MPa. After a grain refining process, the average grain diameter was 1.44×10^{-2} mm and the yield strength increased to 190 MPa. What average grain diameter should be achieved if the desired yield strength is 210 MPa? (*Hint: Use Hall-Petch relation* $\sigma_{ys} = \sigma_0 + k/\sqrt{d}$)	**3**	**3**	**10**
	b.	Explain the mechanism of strain hardening with diagrams. *[Hint: Describe dislocation generation through Frank-Read source]*	**3**	**2**	**10**
5.	**a.**	Explain the drivers for use of Ti-alloys in aircraft structures (That is what properties of Ti-alloys promote their use in aircraft structures) Also explain why the use of Ti-alloys may be limited	**4**	**2**	**10**
	b.	What strengthening mechanisms may be used for non-age-hardening Al-alloys? Describe briefly any two of them	**4**	**2**	**10**
		OR			
6.	**a.**	Answer the following questions in brief. Each question carries 2 marks **i.** State two Al-alloy systems which are not age-hardening. **ii.** What are two most important reasons which prompt the use of Ti-alloys in airframe? **iii.** An Al-Cu alloy is solution heat-treated and naturally aged. Which temper designations would be appropriate to designate this treatment? **iv.** Often, Al-Zn alloys are overaged. What is the most important reason for this? **v.** (v) Why do cast Al-alloys not find much use in aircraft structures?	**4**	**2**	 **2** **2** **2** **2** **2**

	b.	Differentiate between **i.** Corrosion and Stress Corrosion **ii.** Alpha-Ti alloys and beta-Ti alloys **iii.** Peak aged alloy and Overaged alloy	**4**	**2**	 **4** **3** **3**
7.	**a.**	There are several advantages of using carbon fibre composites for primary aircraft structures. Explain them. Which ones you think are the top-3 reasons for their use?	**5**	**2**	**10**
	b.	Answer the following questions in brief. Each question carries 2 marks. **i.** If we denote C=carbonization, G=Graphitization and S=Stabilization, what is the correct sequence of these processes in carbon fibre production from PAN fibre? **ii.** Why is E-glass fibre prone to fatigue damage? **iii.** Consider four fibres: (i) Aramid fibre Kevlar49 (denoted by A) (ii) Carbon fibre T300 (denoted by C), (iii) E-Glass fibre (denoted by G), and (iv) Jute fiber (denoted by J). Rank them in the order of their elastic modulus (lowest to highest). **iv.** An aircraft component has an aerodynamic surface which is doubly curved. It is to be made in composite material. Which would be the preferred weave pattern for good drapeability if a fabric is to be used? **v.** Consider the terms "Crystallinity" and "Cross-link density" in reference to the molecular chain structure of polymers. State which one of them, or both of them, are relevant while discussing the processing of thermoset and thermoplastic polymers?	**6**	**2**	 **2** **2** **2** **2** **2**
		OR			
8.	**a.**	The material of a tension member is to be changed from a Ti-6A1-4V titanium alloy to a unidirectional composite of continuous GY-70 carbon fibre in epoxy matrix. The carbon fibre and the epoxy follow a completely linear stress–strain curve. Calculate the volume fraction of GY-70 fibres required in the composite member to match the modulus of the titanium alloy. In addition, estimate the tensile strength of the composite link and compare its strength–weight ratio with that of the titanium alloy. Following data may be used if required Ti-alloy: E=114 GPa, YS=830 MPa, Density=4.50 g/cm^3 GY70 fibre: E=480 GPa, Failure strain=0.38%; Density=1.96 g/cm^3 Epoxy resin: E= 2 GPa, Failure strain = 2%, Density = 1.3 g/cm^3	**5**	**3**	**10**

	b.	Explain why short beam shear test can be used to estimate interlaminar shear strength (ILSS) even though the shear force in the test is normal to the plies. Then, describe the short beam shear test to find ILSS of a laminated composite.	**6**	**2**	**10**